

TIANYU (TERRY) ZHANG

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EDUCATION

The University of Texas at Austin <i>Austin, Texas</i>	Bachelor of Science in Mechanical Engineering Minor: Computer Science Overall GPA: 3.8	May 2025
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Relevant Coursework: Data Structures and Algorithms, Databases, Data Analytics, Web Development, iOS Programming, Mechatronics, Dynamic Controls & Systems, Heat Transfer, Solids, Statics & Dynamics

Skills: SolidWorks, Google Cloud Database, Python, MATLAB, Java, JavaScript, UNIX, HTML, CSS, SQL, Swift

PROJECTS

CS 329E Mobile Computing - Automated Plant Care Taker App	August 2024 - December 2024
• Programmed an Arduino Uno R4 to interface with light, temperature, and soil moisture sensors using C/C++ • Built an iOS app in Swift with a Bluetooth interface to display real-time sensor data and send control signals • Integrated mobile and embedded systems, managing BLE protocols and hardware timing for intercommunication	
CS 327E Web Development - Personal Website	August 2024
• Designed and developed a personal portfolio website using HTML and CSS to showcase projects and experience • Implemented responsive web design principles to ensure optimal performance and layout across all devices • Deployed the website using AWS S3 and CloudFront, configuring static website hosting and content distribution	
CS 329E Data Analytics - <i>Data Analyst</i>	August 2024
• Utilized SVM modeling to analyze a student database, aiming to predict grades from factors outside of school • Processed and cleaned a dataset of 2313 students using pandas, ensuring data provides accurate analysis • Performed feature engineering to identify predictors like parental support, improving model performance to 83%	
ME 338 Machine Elements - <i>Collaborative RC Car Build</i>	July 2024 - August 2024
• Collaborated with a team of 2 to build a radio controlled car given a battery, controller, receiver, and motor • Developed a scaled down Ackerman steering bracket and modeled the assembly in SolidWorks for 3D printing • Conducted design of experiments, optimizing turning radius and top speed, giving 25% reduction in lap times	

EXPERIENCE

Texas A&M University - <i>Undergraduate Research Assistant</i> ; College Station, TX	August 2021 - May 2022
• Worked 10 hours weekly in a chemistry lab researching how specific nanoparticles affect wastewater filtration • Designed a system for crossflow filtration using a peristaltic pump for testing of membranes made in the lab • Conducted electrospinning experiments to synthesize lignin membranes with graphene oxide nanoparticles	
SpiderSmart Learning Center - <i>Algebra 2 Tutor</i> ; Remote	January 2022 - August 2022
• Prepared custom algebra 2 lessons for 10 students resulting in all students achieving 90+ points in the course • Formatted weekly problem sets through Latex and assessed each student's individual homework performance • Tracked each student's progress using spreadsheets to identify strengths and areas needing improvement	

LEADERSHIP AND ACTIVITIES

Student Engineers Educating Kids (SEEK) - <i>VP Internal Relations, Program Officer</i>	August 2022 - May 2025
• Introduced STEM concepts to elementary students through 10 different engineering projects every semester • Led 20 mentors to demonstrate an engineering project for kids effectively and efficiently over a 12 week period • Created a mechanical engineering project for students aged 6-14 by using real world examples and analogies • Planned biweekly social and sport events for over 200 members fostering greater participation and networking	
Texas Men's Club Volleyball - <i>Libero</i>	January 2024 - May 2025
• Committed 4.5 hours per week to rigorous training and 4 weekends to full day tournaments each season • Developed leadership skills by coordinating defensive plays and supporting teammates on and off the court	

ADDITIONAL INFORMATION

Languages: Fluent in English and Conversational Mandarin Chinese

Interests: Volleyball, Cooking, Baking, Flute, Guitar, Piano, Aviation

Work Eligibility: Eligible to work in the U.S. with no restrictions